Integral modular Hermite accumulator

November9,2026. Mathematical source `15ab03e54`. Unsigned producer evidence throughout [pcc:unverified].

Construction. For the actual source grid and orders, put

$$K = 8h2^a + 8M(a+1), \qquad h = \lceil 3M/u \rceil.$$

Every actual node's child-precision overhead is at most K. Applying the existing coefficient-error theorem at output precision zero, then multiplying by the remaining nonnegative power of two, establishes

$$g_{t,j} = 2^K E_t(0)(-t/\alpha)^j \in \mathcal{O} \quad (j < q_t).$$

Here $\mathcal{O}$ is the concrete rational two-adic integral subring. Define

$$G(J) = \sum_t \sum_{j<q_t} g_{t,j} J_{t,j} \in \mathcal{O}.$$

Lean identifies its rational value with 2^K times the actual Hermite recovery functional for the prescribed quotient degrees.

Cross-precision transition. The actual residue kernel is (2^p). Divisibility cancellation in the integral domain therefore gives

$$[2^K x]_{p+K} = [2^K y]_{p+K} \iff [x]_p = [y]_p.$$

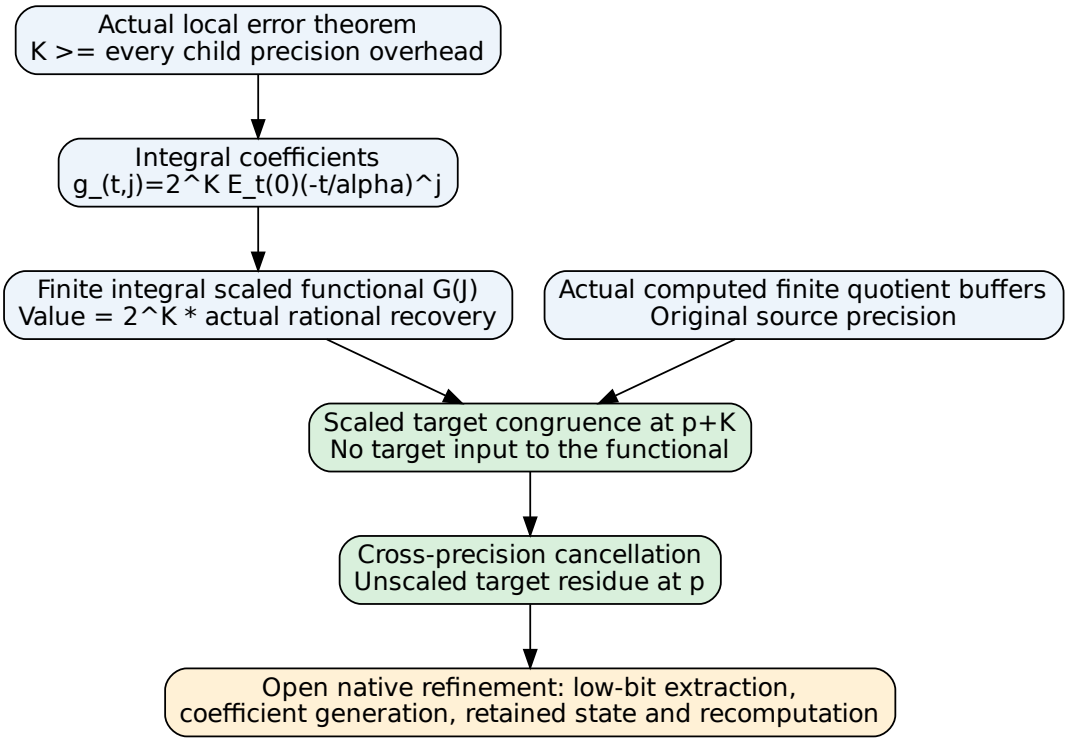
No inverse of two in a fixed finite-characteristic ring is assumed. For the actual computed finite quotient buffers, the scaled accumulator has

$$[G(J)]_{p+K} = \left[2^K \sum_x A_x B_x \right]_{p+K}.$$

The target is not an input to the functional. Domain: integral rational weights, $3 \le u \le a$, $u \le M$, injective supplied placement. p is bit precision.

Representation scope. A finite sum admits fold evaluation, so its algebra does not require retaining every node's quotient vector simultaneously. Native low-bit extraction, efficient coefficient/input-jet construction, peak retained state and recomputation costs remain unproved. This is an algebraic streaming bridge, not measured memory savings or a recursive cost theorem.

Dependencies and checks



Five canonical unsigned PCC bindings (prefix `PCC-20261109-`): residue equivalence9fdb17; cross-precision cancellation445413; actual scaled coefficient integrality d2a76b; scaled-functional mapping f27d83; actual computed scaled target residue eb2e7e. Canonical hashes, successful and failed build logs, and final audit/replay receipts are retained beside the Lean sources and linked in the companion Markdown report.

Sources: actual residue kernel, actual Hermite coefficient-error theorem, constructed finite recovery and the pinned manuscript D.9/D.10 cited in the preceding report. KB unavailable; no novelty assertion. A native accumulator arm requires a prospective amendment to the currently frozen census. No census, timing or memory measurement is reported in this phase.